## Azure SQL Hyperscale Unveiled

As organizations build increasingly data-driven and AI-powered applications, they need database platforms that can grow without sacrificing performance or resiliency. In this practical guide, Bob Ward, a Principal Architect on the Microsoft Azure Data team, takes readers inside Hyperscale's distributed architecture, showing how it uses the power of SQL Server to deliver the scale, performance, and resiliency required by modern cloud workloads.

You will learn how to deploy Hyperscale, build applications that use JSON, vectors, in-database RAG, AI agents, and GitHub Copilot, and leverage core capabilities such as serverless compute autoscaling, elastic pools, automatic storage growth, high availability and disaster recovery, named replicas, and zone redundancy. Practical design patterns demonstrate how teams can start small, scale as needed, and combine familiar SQL Server skills with new cloud-first, AI-ready techniques.

Whether you are modernizing existing applications or building new intelligent systems, this guide equips you to use Azure SQL Hyperscale as a flexible, reliable, AI-enabled cloud database platform for workloads at any scale.

**What You Will Learn**

- Deploy Azure SQL Hyperscale for workloads of any size
- Translate existing SQL Server skills to cloud-based development
- Understand the distributed architecture and internals of Hyperscale
- Build data-driven applications using JSON, vectors, RAG, and AI agents
- Develop Hyperscale solutions using GitHub Copilot and related tools
- Secure cloud data and implement high availability and disaster recovery strategies
- Optimize performance across Hyperscale workloads
- Apply practical design patterns for scalability, resiliency, and modernization

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**Shelve in:**
Databases/Microsoft

**User Level:**
Intermediate–Advanced

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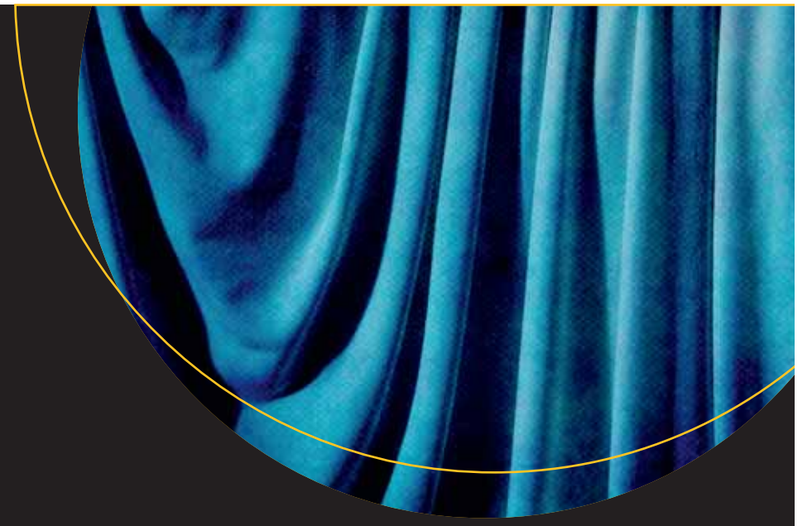

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The Cloud SQL AI Database for Everyone

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Bob Ward

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